deployment depends on more than successful installation. Products must continue operating throughout their expected service life, which increasingly extends to five years or more. Lifecycle planning includes commissioning, software maintenance, firmware upgrades, diagnostics, and long-term technical support. Establishing these processes before large-scale deployment improves long-term maintainability.

Power quality and communication reliability have a direct impact on the performance of IoT devices in large-scale deployments. While laboratory testing is usually carried out under stable conditions, field installations are exposed to wide voltage fluctuations, electrical disturbances, and inconsistent communication networks.

Street lighting automation demonstrates why power quality must be considered during product design. Field installations may experience significant voltage fluctuations and electrical disturbances that are rarely replicated in laboratory environments.

Communication architecture requires similar consideration. Dependence on a single communication path without adequate recovery mechanisms can increase operational risk. Local service interfaces such as Bluetooth or Wi-Fi can allow technicians to access devices, retrieve logs, and perform diagnostics when the primary wide-area network is unavailable. Where the use case requires it, alternate communication paths, retry logic, and store-and-forward capabilities can further improve resilience.

5. Scaling the process, not just the product. Scalability depends not only on the product but also on the processes that support its deployment and long-term operation. Pilot projects often succeed because they involve close supervision, manual intervention, and a limited number of variables. As deployments expand from a handful of devices to thousands, those assumptions no longer hold, making well-defined operational processes just as important as robust product design.

Scalable deployments rely on disciplined workflows established during the design phase rather than after products enter the field. Installation, onboarding, commissioning, diagnostics, reporting, and maintenance should follow consistent procedures that remain effective regardless of deployment size. Version control, asset mapping, issue escalation, and reporting consistency further ensure that engineering teams, service personnel, and customers work within the same operational framework.

A structured deployment process also improves long-term maintainability. Standardised onboarding, software management, and diagnostic procedures enable organisations to monitor deployed devices more effectively, resolve issues faster, and support projects ranging from a few installations to hundreds of thousands. In large-scale IoT deployments, scalability reflects not only product maturity but also the maturity of the processes supporting it.

Keeping the customer at the centre of product design

Successful IoT products are defined not only by innovative technology but also by their ability to deliver dependable performance throughout their operational life. Engineering decisions should prioritise functionality that delivers measurable value while ensuring products remain practical to deploy, operate, and maintain.

Ease of installation and operation is equally important for successful large-scale deployments. Most users expect products that can be deployed quickly with minimal technical assistance. Streamlined onboarding, intuitive interfaces, and straightforward maintenance simplify large-scale deployments, while built-in diagnostic capabilities enable technicians to commission devices, retrieve logs, and perform routine servicing efficiently. These features reduce downtime, lower support costs, and improve long-term maintainability.

Building a successful IoT product involves far more than developing a functional prototype. Products intended for large-scale deployment must be engineered for real operating conditions, simple installation, remote diagnostics, lifecycle management, and consistent service processes. Ultimately, IoT scale is measured not only by the number of connected devices—it is measured also by whether those devices continue delivering dependable value throughout their operational life. EFY

Routine activities such as commissioning, diagnostics, troubleshooting, and software updates are most effective when they can be performed without support from development teams. Built-in diagnostic capabilities can simplify field maintenance. Instead of relying on specialised debugging hardware, products can use integrated communication technologies such as Bluetooth to allow technicians to connect using a smartphone, retrieve diagnostic logs, and upload them directly to cloud-based support systems. This approach enables engineering teams to analyse field issues remotely while reducing the time required to identify and resolve faults.

Over-the-air firmware updates are one example of lifecycle planning. Updating thousands of deployed devices individually is impractical, making remote software management essential for large IoT installations. Version control, reporting mechanisms, and maintenance procedures should also be standardised so that the same operational process can support projects ranging from a few devices to hundreds of thousands.

This article is based on the talk, 'Key Mistakes and Lessons Learned While Designing and Scaling Real-World IoT & AIoT Products' by Mandar Halbhavi, Deputy General Manager, Jio, an IoT and AIoT strategy and solutions professional with more than 18 years of experience across product engineering, smart infrastructure, connected devices, business strategy, and large-scale technology deployments. It has been adapted for publication by Ananthu Ashok, Electronics For You.